This study evaluated several methods of propensity score analysis in a modestly high dimensional setting (142 predictors, 48 being uninformative). My main concerns are the following:

1. Several "estimators" are described on page 5 through 6, however it is unclear if these different methods are being used to perform variable selection, propensity score estimation, or both. In fact, in its present form, the article does not specify what the causal estimand is, why propensity scores are needed for its estimation, how the propensity scores were calculated, or how the propensity scores were ultimately used (e.g. inverse-probability weighting, matching, subclassification, etc). These points all need to be clarified to enable readers to interpret the simulation studies.
2. It is unclear how the error bars presented in the figures were calculated, and consequently whether uncertainty is being appropriately quantified. There should be an association between the number of variables included in the propensity score model and the variability of the final estimator, with models tending to select more variables having greater variability. If this expected trend is present, it is not apparent from the figures. Most concerning are the risk difference estimates in Figure 3 (right), where multiple methods selecting meaningfully different numbers of covariates (Genetic algorithms [64] through XGBoost [48]) apparently result in nearly identical point estimates and confidence intervals. This result needs to be double checked.

Following are some additional suggestions:

1. Please provide a brief overview of "plasmode simulation" in the introduction for readers who are unfamiliar, clarifying how this differs from a standard simulation.
2. Within the introduction, please explain what a propensity score is, why it is needed, and how it is used.
3. Please elaborate on the calculation of the "comorbidity burden measure".
4. The Methods should include a clear explanation of the target causal estimand (i.e. what parameter is being estimated) and how the propensity score is used to estimate that quantity (inverse-probability weighting, atmching, subclassification etc). In addition, please clarify whether the machine learning models in the "Estimators" section are being used for variable selection, propensity score estimation, or both.
5. Additional context is needed in Figure 1: what parameter is being estimated? The risk difference, the odds ratio, the true propensity score? What is the ground truth against which the bias is being calculated?
6. In addition to bias and coverage probability, it would be useful to examine the true positive rate (probability of selecting a truly associated covariate) and the false positive rate (probability of selecting a noise covariate).

7. Figure 3 is concerning. The kitchen sink model, which includes all predictors, should be low bias but high variance, yet its confidence intervals are not appreciably larger than those of the other methods. For the risk difference, please explain why some methods are producing asymmetrical confidence intervals. In addition, it seems unlikely that 6 methods, Genetic Algorithm through XGBoost, would select different numbers of features yet produce nearly identical point estimates and confidence intervals for the risk difference. Please explain. Further, how are these methods yielding different estimates for the odds ratio yet the same estimates for the risk difference?